

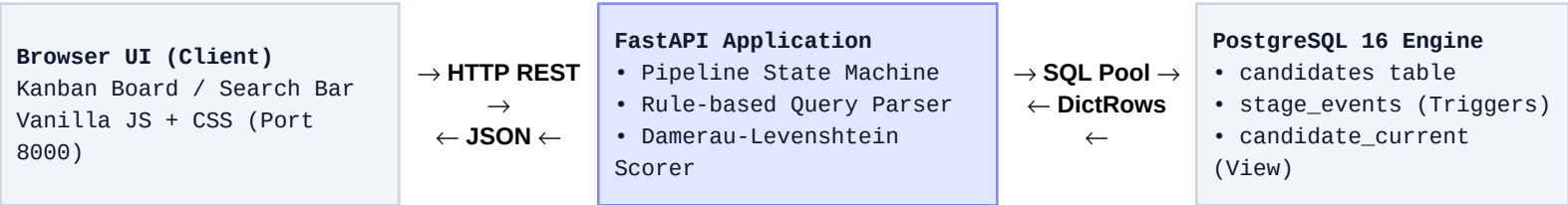
Mini Hiring Pipeline — Architecture & Decisions

GitHub Repository: <https://github.com/RohitPawar001/hiring-pipeline>

Stack: FastAPI (Python 3.12) | PostgreSQL 16 | Docker Compose | Vanilla JS

System Architecture

The system is designed with a strict separation between state validation, event-sourced audit persistence, and natural language search parsing:



Key Engineering Decisions

- **1. Event-Sourced Audit Trail & Immutability:** Rather than mutating candidate status in-place, every transition produces an immutable record in `stage_events`. Database triggers (`no_update_delete` and `no_truncate`) enforce append-only guarantees at the engine level via `PL/pgSQL forbid_change()`, ensuring audit compliance.
- **2. Derived Current Stage:** Current stage is never stored as a mutable column. Instead, the `candidate_current` view dynamically computes state using a `JOIN LATERAL ... ORDER BY id DESC LIMIT 1` query, eliminating state-history drift.
- **3. Concurrency Protection:** Stage transitions use `SELECT 1 FROM candidates WHERE id=%s FOR UPDATE`. This row-level lock prevents race conditions and simultaneous invalid stage transitions.
- **4. Deterministic Rule-Based Search:** A structured regex parser with stopwords filtering tokenizes natural language queries without unpredictable LLM latency, cost, or hallucination. It provides clear, actionable 422 error messages on invalid input.
- **5. Damerau-Levenshtein Fuzzy Ranking:** Tailored string matching correctly penalizes single-character transpositions (e.g. `'sharam' → 'sharma' = 1` edit distance) unlike standard Levenshtein (2 edits), yielding intuitive typo-tolerance.
- **6. Active-Candidate Duration Semantics:** Queries specifying stage duration (e.g. `'stuck in Screening'`) automatically filter for active candidates and exclude terminal outcomes (Hired / Rejected).

Engineering Judgment: Disagreeing with AI Suggestions

AI Suggestion: The initial test teardown proposal suggested running `TRUNCATE candidates CASCADE` between tests.

Why I Disagreed: Our PostgreSQL schema attaches a strict `BEFORE TRUNCATE` trigger on `stage_events` that raises an exception. A cascading truncate would fail immediately because the engine protects audit records.

Resolution: Decoupled tests by assigning unique `UUID` prefixes to candidate names for each test case, preserving audit immutability without violating DB trigger rules.